Conserve mechanical energy

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Track kinetic and potential energy when nonconservative work is negligible.

SUCCESS CRITERIA - I can...

1. Identify the energy system and assumptions.
2. Keep the total energy consistent between positions.
3. Explain how height and speed trade off.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “mechanical energy” and “conserved” to explain your model or result. Sentence starters are optional.

mechanical energy: total kinetic and potential energy in a model

conserved: unchanged in total under the stated conditions

system: the object or objects chosen for analysis

01 Bring back an earlier idea

Lift a 3 kg object by 5 m.
Use g=10 N/kg.
Find the change in gravitational potential energy.

Word help: potential energy = energy associated with the configuration of interacting objects

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $K_i + U_i = K_f + U_f$; $K = \frac{1}{2}mv^2$; $U = mgh$.
Mechanical energy conserved when only conservative interactions transfer energy. Use g = 10 N/kg near the surface.

Worked reasoning: A 1 kg object drops 5 m from rest with g=10 m/s² and no energy loss. It loses 50 J of potential energy and gains 50 J of kinetic energy, giving speed 10 m/s.

Your turn: Draw energy bars at the start and just before arrival.
Label the system and neglected losses.

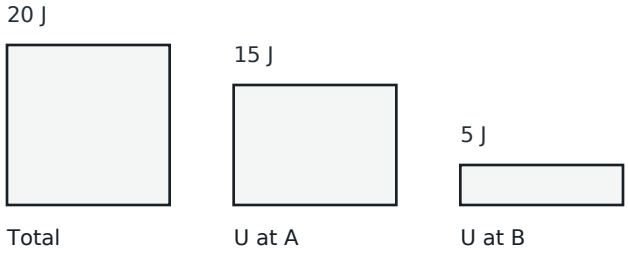
Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: ① ② ③

03 Look and explain

A system has total mechanical energy 20 J. At one point U=15 J; at another U=5 J.
Find K at each point.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Bar height represents energy. Read the value and unit above each bar.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

04 Build the solution

A 1 kg object starts at rest 2 m above the zero level and falls ideally.
Find starting U and final K at zero height.

Word help: at rest = not moving in the chosen reference frame

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: 1 2 3

05 Use the idea

A cart loses mechanical energy on a rough track. Does total energy conservation fail?
Name the destination to include.

Sentence starter: I notice ... . This supports ... because ... .

Marzano check - circle: 1 2 3

06 Choose a method

An impactor has 4 times the original mass at the same speed and stopping time.
Compare momentum and average stopping-force size.

Word help: momentum = mass times velocity, including direction

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: 1 2 3

07 Exit check

An ideal system has 30 J total. At a point U=12 J, what is K?

Word help: ideal = using stated simplifying conditions

Sentence starter: I notice ... . This supports ... because ... .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Two frictionless ramps start at the same height and end at the same lower height.
Compare final speeds and explain what need not be equal.

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: 1 2 3